

PORT TOWNSEND HARBOR, WA, USA

WMO: 994025

Lat: 48.1110N

Lon: 122.7600W

Elev: 16

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	28.5	32.2	N/A	N/A	N/A	N/A	N/A	N/A	32.5	46.5	28.2	46.9	7.5	310	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.3	76.8	N/A	72.7	N/A	69.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.4	22.3	18.5	DB	25.2	82.9	3.0	2.5	23.0	84.7	21.3	86.2	19.6	87.7	17.4	89.5	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	43.7	42.8	45.9	49.2	54.0	57.7	60.9	61.6	58.7	52.5	46.2	42.5
	DBStd	7.97	4.15	5.01	4.15	3.24	3.84	3.52	3.58	3.17	3.15	3.57	4.90	5.03
	HDD50	977	197	204	137	53	7	0	0	0	0	16	129	234
	HDD65	5003	660	623	591	475	340	223	138	115	189	387	565	697
	CDD50	1473	2	1	11	27	132	231	338	360	262	94	13	2
	CDD65	26	0	0	0	0	0	4	11	10	1	0	0	0
	CDH74	218	0	0	0	0	2	35	83	86	12	0	0	0
	CDH80	21	0	0	0	0	0	3	11	7	0	0	0	0
Wind	WSAvg	6.9	7.1	7.6	8.0	7.7	6.4	6.4	6.0	5.4	5.0	6.7	8.5	8.7
Precipitation	PrecAvg	19.2	2.2	1.6	1.7	1.5	1.2	0.9	0.9	1.1	1.4	2.5	2.6	
	PrecMax	24.3	6.6	3.8	3.5	3.4	3.0	3.1	2.9	2.8	3.2	4.4	4.8	5.3
	PrecMin	13.5	0.2	0.3	0.3	0.3	0.3	0.2	0.0	0.0	0.1	0.0	0.6	0.4
	PrecStd	3.0	1.2	0.8	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.9	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.2	54.6	60.0	64.1	72.1	78.5	81.5	80.4	75.5	65.9	58.0	55.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	52.1	52.1	56.2	59.7	67.6	71.8	76.5	76.6	71.4	61.9	55.5	51.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	50.5	50.3	53.5	56.8	63.5	67.4	72.0	71.7	68.0	59.4	53.8	50.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	49.0	48.8	51.6	54.3	60.3	64.1	68.5	68.7	64.8	57.5	52.1	48.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	5.9	6.7	8.7	9.6	11.1	11.5	14.0	14.3	12.0	8.1	6.8	5.6
		MCDBR	7.6	8.2	11.5	13.6	17.0	18.3	21.2	20.0	17.8	10.8	7.6	6.7
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.320	0.316	0.327	0.348	0.374	0.357	0.347	0.351	0.339	0.328	0.323	0.317	
	taud	2.517	2.551	2.536	2.466	2.411	2.480	2.539	2.526	2.555	2.575	2.525	2.478	
	Ebn at Noon	241	271	285	287	282	287	289	283	277	262	237	225	
	Edh at Noon	20	23	27	32	36	33	31	30	27	23	19	18	
All-Sky Solar Radiation	RadAvg	303	581	895	1339	1705	1864	2052	1739	1249	692	360	245	
	RadStd	42	78	105	125	136	182	148	114	108	75	52	34	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page